

Week 8 Lecture Plan

Midterm Debugging: Evidence, Mechanism, Fix, Cost, and Kafka Preview

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Pre-Class Preparation

- ☐ Week 8 lab compiles: `week8-lab/week8-day1-midterm-debugging-kafka-preview`
- ☐ Smoke test passes: `gradlew.bat test -tests "edu.lu.concurrency.week8.day1.DemoSmokeTest"`
- ☐ Instructor check passes: `gradlew.bat -Pinstructor=true instructorCheck`
- ☐ Lecture PDFs ready: `Lecture_08_01_nar.pdf` through `Lecture_08_04_nar.pdf`
- ☐ Lab manual ready: `Lab_Week08_MidtermDebuggingKafkaPreview.pdf`
- ☐ Open `MIDTERM_PACKET.md`, `DIAGNOSIS_REPORT.md`, and `RECORDED_EXPLANATION_RUBRIC.md`

Learning Targets

By the end of Week 8, students must be able to:

1. Read a thread-dump clue and connect it to a concrete async failure.
2. Separate symptoms from mechanism in a diagnosis report.
3. Fix a never-completing async request using timeout and fallback.
4. Explain the performance and correctness cost of the fix.
5. Present a five-minute technical diagnosis to an engineering audience.
6. Explain Kafka's log mental model: history, replay, recovery, audit, and decoupling.

Lecture Structure

Segment	Material	Goal
Part 1	<code>Lecture_08_01_nar</code>	Midterm framing, thread-dump evidence, diagnosis structure
Part 2	<code>Lecture_08_02_nar</code>	Broken async server, timeout/fallback fix, mechanism-first debugging
Part 3	<code>Lecture_08_03_nar</code>	Performance tradeoff, measurement plan, recorded explanation rubric
Part 4	<code>Lecture_08_04_nar</code>	Kafka mental-model preview and Phase 2 bridge

Recommended Time Budget

Time	Activity	Notes
0–10 min	Midterm framing	Rubric: real bug, mechanism, cost.
10–35 min	Part 1 slides and Demo01–Demo02	Read thread-dump evidence and model report.
35–65 min	Part 2 slides and Demo03	Broken async handler and timeout/fallback fix.
65–90 min	Part 3 slides and Demo04	Tradeoff analysis and measurement expectations.
90–110 min	Part 4 slides and Demo05	Kafka preview: logs as ordered history.
110–120 min	Submission briefing	Deliverables and recorded explanation expectations.

Student Deliverables

- Thread-dump evidence or simulated thread-dump interpretation.
- Diagnosis report.
- Fixed async handler.
- Performance tradeoff analysis.
- Five-minute recorded technical explanation.
- Kafka preview answer.

Exit Ticket

1. What is the mechanism behind the broken async server?
2. What evidence supports your diagnosis?
3. What cost does the timeout/fallback fix introduce?